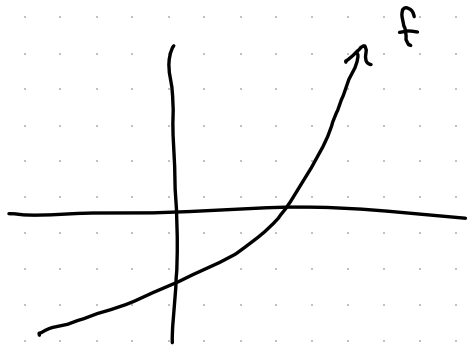


Derivatives of inverse functions - 0.3.7



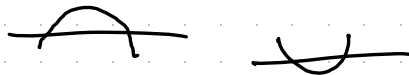
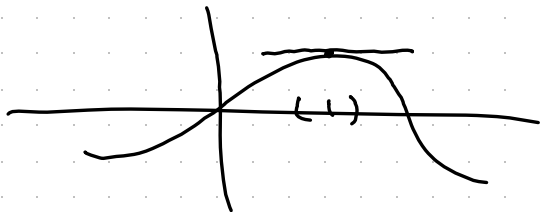
This function is one-to-one,
hence invertible.

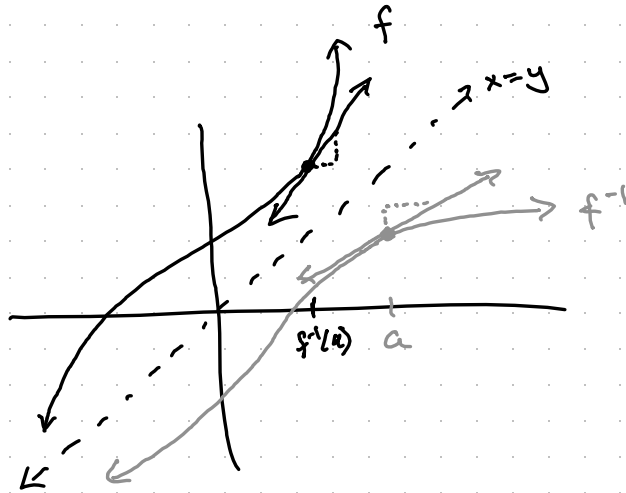
Observation: Increasing functions
are always invertible.

If we don't see the graph of f , we
can take a derivative!

→ if f' is positive, f is increasing,

→ if f' is negative f is decreasing.





Interested in $(a, f^{-1}(a))$
 $\rightarrow$ want to know $(f^{-1})'(a)$.

$$\Leftrightarrow \frac{\text{run}}{\text{rise}} = \frac{1}{f'(f^{-1}(a))}$$

slope at $(f^{-1}(a), a)$ is $\frac{\text{rise}}{\text{run}} = f'(f^{-1}(a))$

slope at $(a, f^{-1}(a))$ is $\frac{\text{run}}{\text{rise}} = (f^{-1})'(a)$

Relationship: these are reciprocals!

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

Another way to get the formula:

$$f(f^{-1}(x)) = x \quad \text{and} \quad f^{-1}(f(x)) = x$$

$$f'(f^{-1}(x)) \cdot \underbrace{(f^{-1})'(x)} = 1 \quad \Leftrightarrow \quad \boxed{(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}}.$$

Example: Find $(f^{-1})'(2)$ for $f(x) = x^5 + x$.

Problem: no nice formula for f^{-1} .

$$(f^{-1})'(2) = \frac{1}{f'(f^{-1}(2))}$$

$$= \frac{1}{5(f^{-1}(2))^4 + 1}$$

$$= \frac{1}{5(1)^4 + 1} = \boxed{\frac{1}{6}}$$

$$f'(x) = 5x^4 + 1 > 0$$

Need to know $f^{-1}(2)$ to use the formula!

Can we find an input to f that gives 2 as an output?

$$f(1) = 2, \text{ thus } f^{-1}(2) = 1.$$